Supplement: Model-Bounded Variational and Expected Free Energy Examples

Supplement: Model-Bounded Variational and Expected Free Energy Examples I

This supplement carries the kernel-level mathematical walkthroughs that support the curriculum without crowding the main Results section. The purpose is narrow: show how the claim registry, figures and prose jointly teach the two optimization objectives that organize the course. Variational free energy is used for belief updating under a fixed generative model; expected free energy is used for policy evaluation under uncertainty. Both examples are deliberately small enough to audit by rerunning the kernels.

Supplement: Model-Bounded Variational and Expected Free Energy Examples I

Notation is fixed before the examples: s is the latent state, o is the observation, $p(s)$ is the prior, $p(o \mid s)$ is the likelihood, $p(o, s)$ is the joint generative density, and $q(s)$ is the recognition density. The variational free-energy convention is $F[q] = \mathbb{E}_{q(s)}[\ln q(s) - \ln p(o, s)] = D_{\mathrm{KL}}(q(s) \| p(s \mid o)) - \ln p(o)$, so the bound is tight when $q(s)$ equals the exact posterior. For expected free energy, this supplement uses the lower-is-better policy convention $q(\pi) = \mathrm{softmax}(-\gamma G(\pi))$ with policy precision γ; the plotted terms are therefore tied to the stated policy set and preference distribution, not to a universal sign convention (Buckley et al. 2017; Parr et al. 2022; Millidge et al. 2021; Champion et al. 2026; Nuijten, Lukashchuk, et al. 2026; Nuijten, Laar, et al. 2026).

SkillTree bridge from derivations to quiz-gated subjects

SkillTree bridge from derivations to quiz-gated subjects I

The supplement is not an isolated appendix of equations. It is a bridge from kernel claims to quiz-gated SkillTree material: `sub_free_energy` and `sub_efe` carry the core VFE/EFE concepts, the scientific-supplement subjects train bound and term interpretation, and the mastery subjects turn those interpretations into route-handoff and quiz-authoring checks. fig. 1 is generated from the current curriculum graph after structural checks and SkillTree route contract so the text, figure, and dashboard artifact all point to the same subject, skill, prerequisite-edge, quiz, and route data. This bridge also keeps the VFE/EFE distinction scholarly rather than mnemonic: VFE is an evidence bound for belief updating, while EFE is a policy-evaluation objective whose formulations and preference assumptions need explicit scope boundaries (Buckley et al. 2017; Parr et al. 2022; Smith et al. 2022; Millidge et al. 2021; Champion et al. 2026; Nuijten, Lukashchuk,

SkillTree bridge from derivations to quiz-gated subjects I

et al. 2026; Nuijten, Laar, et al. 2026).

SkillTree bridge from derivations to quiz-gated subjects II

Layer	Skills	Prerequisite edges	Depth	Representative IDs	Learn route	Quiz route
VFE core	7	16 total; 6 internal	7-10	vm_elbovm_perception_descentvm_message_passing	/subjects/sub_free_energy	/subjects/sub_free_energy/skills/vm_message_passing/quizzes/vm_message_passing_quiz
EFE core	7	15 total; 7 internal	18-20	ai_expected_free_energyai_explore_exploitai_free_energy_of_future	/subjects/sub_efe	/subjects/sub_efe/skills/ai_free_energy_of_future/quizzes/ai_free_energy_of_future_quiz
VFE supplement	6	11 total; 5 internal	9-14	sci_variational_01sci_variational_04sci_variational_06	/subjects/sub_scientific_supplement_variational	/subjects/sub_scientific_supplement_variational/skills/sci_variational_06/quizzes/sci_variational_06_quiz
EFE supplement	6	11 total; 5 internal	19-24	sci_active_inference_01sci_active_inference_04sci_active_inference_06	/subjects/sub_scientific_supplement_active_inference	/subjects/sub_scientific_supplement_active_inference/skills/sci_active_inference_06/

Bridge figure I

Bridge figure I

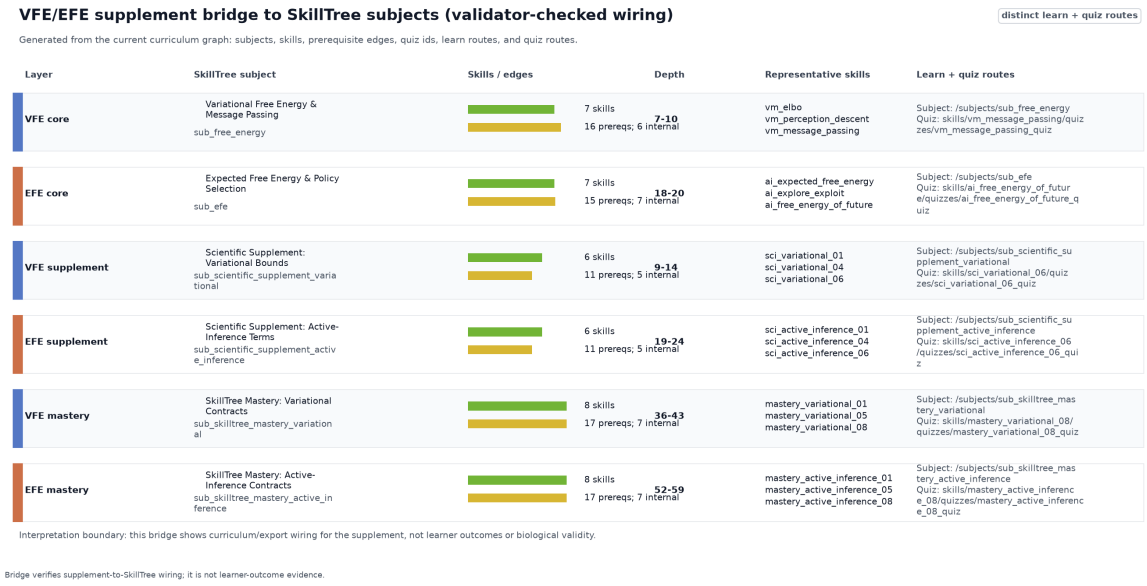


Figure 1: VFE/EFE SkillTree bridge generated from the current curriculum graph after structural checks. Rows are real SkillTree subjects, columns report skill counts, prerequisite-edge counts, graph-depth ranges, representative skill IDs, and learn-route and quiz-route fragments that expose the subject identifiers; color marks separate VFE-centered rows from EFE-centered rows. The figure connects the supplement's mathematical plots to quiz-gated curriculum nodes and prevents the derivation from being presented as a standalone artifact disconnected from the learning path.

Noisy-sensor posterior and the tight variational-free-energy bound

Noisy-sensor posterior and the tight variational-free-energy bound I

On a noisy-sensor model with a 1% base rate, 99% sensitivity, and 95% specificity, a positive observation yields a posterior of 0.166667 for the latent condition. This is the same base-rate calibration problem that makes Bayesian reasoning difficult in natural-language judgment tasks and motivates frequency-format teaching examples for Bayesian inference (Cosmides and Tooby 1996; Gigerenzer and Hoffrage 1995). The evidence probability of the positive observation is 0.0594, so the observation surprise is 2.82346 nats.

Noisy-sensor posterior and the tight variational-free-energy bound I

Noisy-sensor parameter	Paper value
Latent states	`present`, `absent`
Observation index	`positive`
Prior D	$[0.01, 0.99]$
Likelihood rows $A_{o,s}$	`positive`: $[0.99, 0.05]$; `negative`: $[0.01, 0.95]$
VFE sweep coordinate	$q(\text{present})$ over $[0.001, 0.999]$, plus the exact posterior and 0.5

Noisy-sensor posterior and the tight variational-free-energy bound I

Evaluating variational free energy across beliefs confirms the central inequality: a uniform belief gives 3.11735 nats, while the true posterior gives 2.82346 nats, exactly the negative log-evidence. fig. 2 shows the full one-dimensional sweep over $q(\mathrm{present})$: the curve bottoms out at the exact posterior and stays above the surprise line everywhere else. The plot should not be read as a universal geometry of all recognition densities. It is a controlled diagnostic for one discrete model, one observation and one belief coordinate.

Noisy-sensor posterior and the tight variational-free-energy bound I

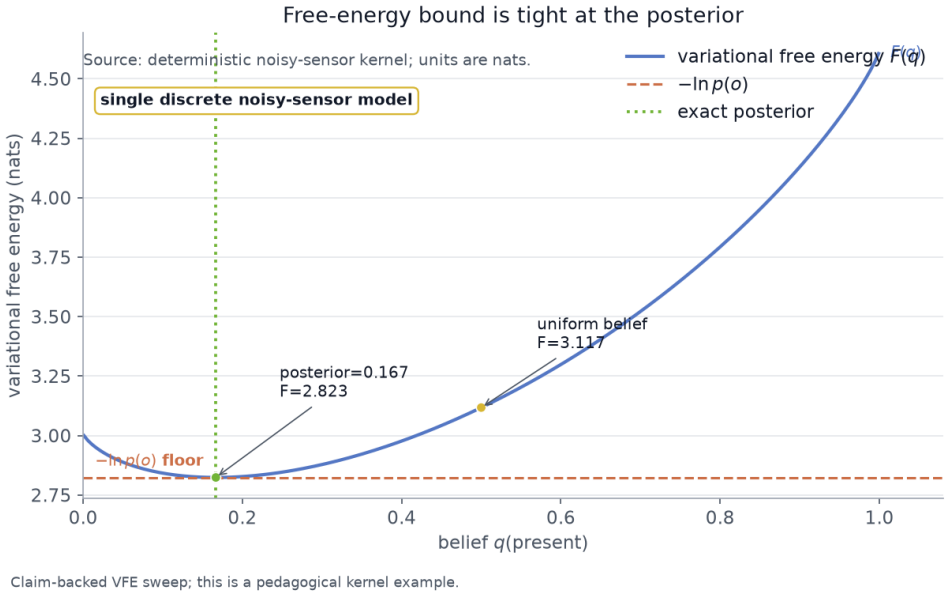


Figure 2: Variational free-energy sweep for the noisy-sensor model, generated from the same claim-registry kernels that provide the manuscript tokens and taught through the VFE core, supplement, and mastery SkillTree subjects in the VFE/EFE SkillTree bridge. The x-axis is the recognition belief $q(\mathrm{present})$, the blue curve is $F(q)$ in nats, the horizontal dashed line is $-\ln p(o)$, the vertical dotted line is the exact posterior, and the labeled points compare the looser uniform belief with the posterior value. The intended takeaway is bound tightness at the posterior for this one discrete model and observation; the plot does not establish that the model is true or that the same geometry holds for every recognition family.

Expected free energy: epistemic value and bounded policy comparison

Expected free energy: epistemic value and bounded policy comparison I

Decomposing expected free energy on an informative sensor shows that an uncertain belief carries positive epistemic value (0.368064 nats) while a certain belief carries none (0 nats): once the state is known, an observation cannot reduce uncertainty. In the two-action agent, preferences are neutral, so the pragmatic term is identical for both actions and the choice is decided by information gain. Under the lower-is-better $G(\pi)$ convention above, the information-seeking *sample* action scores lower expected free energy (0.546837 nats) than *commit* (0.613267 nats), because it carries higher epistemic value (0.146311 nats versus 0.079881 nats). This is the epistemic drive of Active Inference: no separate curiosity bonus is required in the reference formulation used here (Friston et al. 2015, 2017; Da Costa et al. 2020; Sajid et al. 2021). Derivation and unification work is the

reason the supplement keeps the formulation explicit instead of presenting “expected free energy” as a single unqualified object (Millidge et al. 2021; Champion et al. 2026; Nuijten, Lukashchuk, et al. 2026; Nuijten, Laar, et al. 2026).

Preference convention and parameters I

One convention should be made explicit here: the kernel normalizes the preference vector C to a probability distribution before taking its logarithm for the pragmatic term (`preferences = normalize(C)`, then $\mathbb{E}_{q(o)}[\ln \mathrm{normalize}(C)]$), so the values plotted above assume C is a preference *distribution*. This is a specific modeling choice rather than the only option: canonical Active Inference often treats C directly as a vector of log-preferences, which shifts the pragmatic term by an additive constant and can change its scale relative to the epistemic term.

EFE parameter table I

EFE parameter table I

EFE sample/commit parameter	Paper value
Sensor likelihood	$A = [[p, 1-p], [1-p, p]]$, with $p = 0.9$
EFE decomposition preference	$C = [0.9, 0.1]$ for the informative-sensor term example
Agent preference	Neutral $C = [0.5, 0.5]$ for the sample/commit policy comparison
Prior before observation	$D = [0.5, 0.5]$
Actions	`sample` uses identity dynamics; `commit` maps both prior states toward state `0` with transition columns $[0.95, 0.05]$
Policy posterior	$q(\pi) = \mathrm{softmax}(-\gamma G_\pi)$ over the two one-step policies

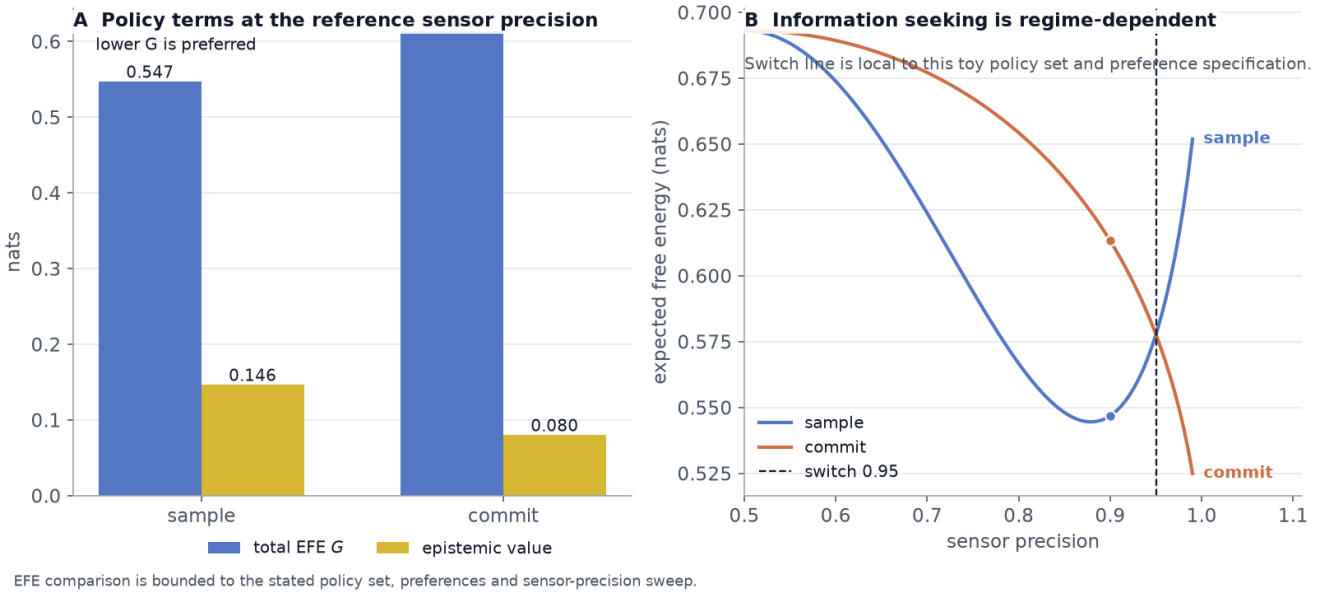
Sensor precision as a sample-versus-commit policy boundary

Sensor precision as a sample-versus-commit policy boundary I

The expected-free-energy result is not a universal law. It is pinned to a specific moderate-precision regime: at sensor precision 0.9, sampling wins; at the computed switch precision 0.95, sample and commit tie; above that point, the commit action can become lower in expected free energy. fig. 3 shows both the sample/commit comparison and the precision sweep, making the regime boundary explicit instead of hiding it in prose (Millidge et al. 2021; Champion et al. 2026; Nuijten, Lukashchuk, et al. 2026; Nuijten, Laar, et al. 2026).

Sensor precision as a sample-versus-commit policy boundary I

Precision-sweep parameter	Paper value
Sensor precision range	$p \in [0.5, 0.99]$ with the computed switch inserted into the grid
Sweep samples	101 grid samples before inserting $p = 0.9$ and the switch precision
Prior, preference, actions	Same neutral two-action agent as the sample/commit comparison
Tie criterion	The switch is the bisection point where $G_{\text{sample}} - G_{\text{commit}} = 0$



EFE comparison is bounded to the stated policy set, preferences and sensor-precision sweep.

Sensor precision as a sample-versus-commit policy boundary II

Precision-sweep parameter	Paper value
Sensor precision range	$p \in [0.5, 0.99]$ with the computed switch inserted into the grid
Sweep samples	101 grid samples before inserting $p = 0.9$ and the switch precision
Prior, preference, actions	Same neutral two-action agent as the sample/commit comparison
Tie criterion	The switch is the bisection point where $G_{\text{sample}} - G_{\text{commit}} = 0$

Figure 3: Expected free-energy decomposition for the sample/commit task, generated from the discrete active-inference kernel and taught through the EFE core, supplement, and mastery SkillTree subjects in the VFE/EFE SkillTree bridge. The left panel compares total expected free energy and epistemic value for the two policies at sensor precision 0.9; the right panel sweeps sensor precision and marks the computed sample/commit switch. The visual encoding keeps policy score and epistemic value separate so the reader can see why sampling is preferred in the reference regime, while the right panel prevents the teaching example from being overgeneralized beyond the stated policy set and preferences.

Sensor precision as a sample-versus-commit policy boundary I

fig. 4 broadens the sensitivity view without changing the evidence class. The left panel varies the base rate in the same noisy-sensor likelihood and shows how a positive observation's posterior changes as the prior changes. The middle panel repeats the VFE bound profile so the equality point remains tied to the exact posterior. The right panel repeats the EFE precision sweep and marks the computed sample/commit boundary. These panels are deterministic kernel diagnostics: they identify where the teaching example is stable or regime-limited, not whether a learner or a biological system will behave that way.

Sensor precision as a sample-versus-commit policy boundary I

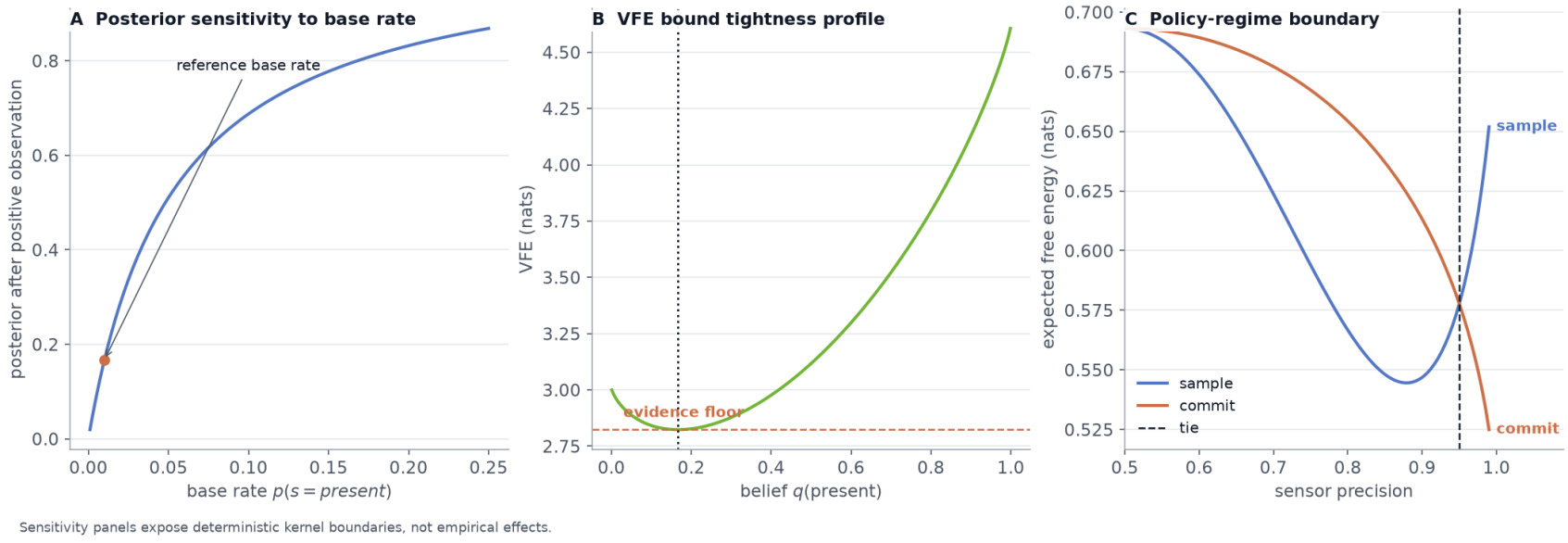


Figure 4: Kernel sensitivity profiles for the supplemental examples, generated from the noisy-sensor posterior, VFE sweep, and EFE precision sweep, then positioned in the SkillTree bridge as a route from mathematical identity to quiz-gated interpretation. The left panel varies the base rate and shows posterior sensitivity after a positive observation, the middle panel plots the VFE bound profile with posterior and evidence references, and the right panel sweeps expected free energy for sample and commit policies across sensor precision while marking the computed policy switch. These panels identify deterministic regime boundaries for teaching examples; they are not empirical estimates, confidence intervals, or learner outcome statistics.

Exact inference on a chain with sum-product belief propagation

Exact inference on a chain with sum-product belief propagation I

The message-passing curriculum is bound to a worked example rather than left as prose. Because the example factor graph is acyclic, sum-product belief propagation returns the exact node marginals for a three-node sticky latent chain with a single informative reading at the middle node. The observed node settles at 0.9, and the smoothing message lifts each unobserved neighbor from its prior to 0.74 – a lift of 0.24 that is produced purely by propagating the one reading along the transition factor. Because a chain is a tree, these marginals are the same numbers a brute-force enumeration of the full joint would give; the test suite pins that equality, so the example is an exact-inference check, not an approximation (Kschischang et al. 2001; Pearl 1988; Parr et al. 2022). Factor-graph message passing is also a live implementation path for EFE minimization, but this supplement uses the chain only to teach exact

Exact inference on a chain with sum-product belief propagation I

sum-product inference and keeps policy-optimization message passing out of scope for this release (Nuijten et al. 2025).

Sum-product parameter table I

Sum-product parameter table I

Sum-product parameter	Paper value
Chain length	3 binary latent nodes
Prior on node 0	$[0.5, 0.5]$
Sticky transition	$[[0.8, 0.2], [0.2, 0.8]]$
Evidence potentials	node 0: $[1, 1]$; node 1: $[0.9, 0.1]$; node 2: $[1, 1]$

Conjugate parameter learning from Dirichlet beliefs

Conjugate parameter learning from Dirichlet beliefs I

Parameter learning is shown as a Dirichlet-conjugate update of one sensor column. A flat prior has mean 0.5; after a fixed observed count vector the posterior mean sharpens to 0.75, and the expected log-parameter that variational message passing actually consumes is -0.30202 nats, computed through the digamma identity $\mathbb{E}[\ln p_i] = \psi(\alpha_i) - \psi(\sum_j \alpha_j)$. This is the learning half of active inference: beliefs about model parameters move with evidence, not just beliefs about states (Da Costa et al. 2020).

Bayesian model reduction: whether data require added structure

Bayesian model reduction: whether data require added structure I

Bayesian model reduction compares the evidence of a full model against a reduced model whose pruned state carries negligible prior mass. After a positive reading on the noisy-sensor model the full-model log evidence is -2.82346 nats and the reduced-model log evidence is -2.99571 nats, so the log Bayes factor is 0.172252 nats. The positive sign says the data support keeping the pruned structure: the reduced model is rejected. Both terms are exact log evidences, so the reported factor is the difference of two computed numbers rather than a hand-typed value (Friston et al. 2018; Parr et al. 2022).

Bayesian model reduction: whether data require added structure I

BMR parameter	Paper value
Full prior	$D = [0.01, 0.99]$
Reduced prior	the `present` state is set to 10^{-6} and the prior vector is renormalized
Observation and likelihood	same positive observation and noisy-sensor likelihood as the VFE example
Score	$\log P(o \mid \text{full}) - \log P(o \mid \text{reduced})$

Ambiguity as a state-dependent component of expected free energy

Ambiguity as a state-dependent component of expected free energy I

Expected free energy carries an ambiguity term – the expected entropy of the likelihood columns – that the main decomposition leaves implicit. On an asymmetric sensor whose two states differ in how noisy their readings are, the crisp state has likelihood entropy 0.325083 nats, while a uniform belief averages to 0.509115 nats. Ambiguity is therefore a property of where the agent believes it is, which is why reducing it is part of what drives exploration (Parr et al. 2022; Sajid et al. 2021).

Multi-step planning and model-dependent look-ahead value

Multi-step planning and model-dependent look-ahead value I

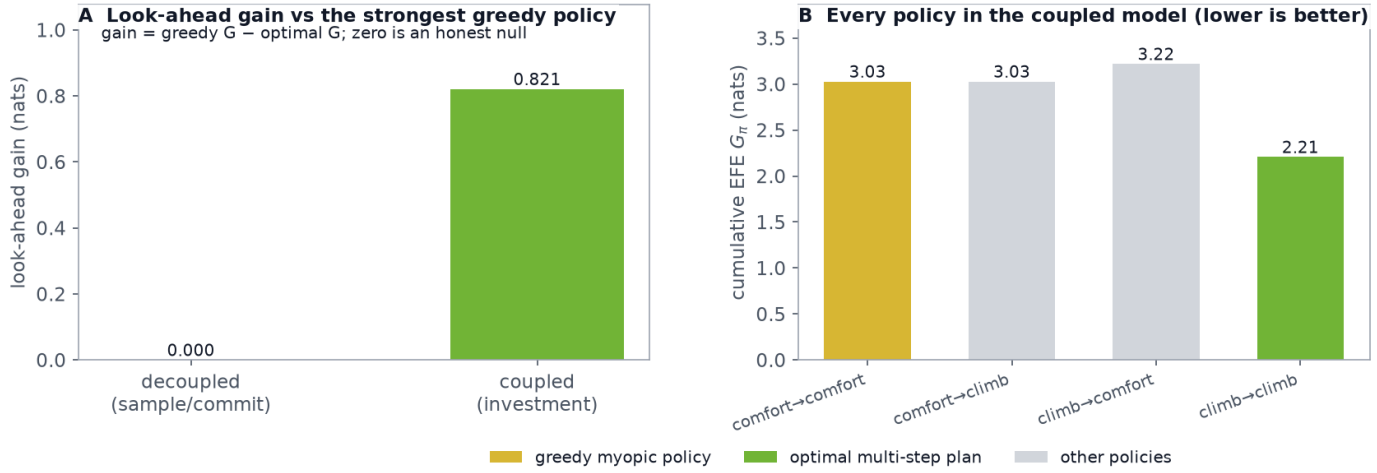
The one-step agent above is generalized to a planner that scores every length-H action sequence by its cumulative expected free energy and forms a posterior over policies, $q(\pi) = \mathrm{softmax}(-\gamma G_\pi)$; at $H = 1$ the per-policy values reduce exactly to the one-step result, a consistency invariant the tests pin. Whether planning beats a myopic agent is then a measured question, not an assumption, and the answer is regime dependent.

Multi-step planning and model-dependent look-ahead value I

In a decoupled sample/commit model the optimal multi-step plan coincides with the strongest greedy policy, so the look-ahead gain is exactly 0 nats – an honest null. In a coupled investment model, where a state that is myopically attractive blocks access to a higher-value goal, the planner pays a worse first step to reach the goal: the greedy policy scores 3.02826 nats, the optimal plan scores 2.20728 nats, and the look-ahead gain is 0.820981 nats with the policy posterior concentrating 0.915376 of its mass on the winning plan (fig. 5). The comparator is the strongest myopic policy, not a weakened straw baseline, so the gain is a fair number; the first-principles lesson is that look-ahead pays only when immediate and long-run value diverge (Parr et al. 2022; Da Costa et al. 2020).

Multi-step planning and model-dependent look-ahead value I

Planning parameter	Decoupled sample/commit model	Coupled investment model
States	2 latent states	`start`, `comfort`, `goal`, `climb`
Observation model	informative sensor with $p = 0.9$	identity likelihood
Preference vector	$C = [0.9, 0.1]$	$C = [0.03, 0.22, 0.55, 0.20]$
Initial prior	$D = [0.5, 0.5]$	$D = [1, 0, 0, 0]$
Actions	`sample` identity; `commit` collapse toward state 0	`comfort` sends start to comfort; `climb` sends start to climb and climb to goal
Horizon and policy precision	$H = 2$, $\gamma = 4$	$H = 2$, $\gamma = 4$



Look-ahead value is a measured, model-dependent quantity from the planning kernel; it is not a universal property of planning.

Multi-step planning and model-dependent look-ahead value II

Planning parameter	Decoupled sample/commit model	Coupled investment model
States	2 latent states	`start`, `comfort`, `goal`, `climb`
Observation model	informative sensor with $p = 0.9$	identity likelihood
Preference vector	$C = [0.9, 0.1]$	$C = [0.03, 0.22, 0.55, 0.20]$
Initial prior	$D = [0.5, 0.5]$	$D = [1, 0, 0, 0]$
Actions	`sample` identity; `commit` collapse toward state 0	`comfort` sends start to comfort; `climb` sends start to climb and climb to goal
Horizon and policy precision	$H = 2$, $\gamma = 4$	$H = 2$, $\gamma = 4$

Figure 5: Multi-step planning look-ahead, generated from the discrete active-inference planning kernel. The left panel contrasts the measured look-ahead gain (optimal multi-step plan versus the strongest greedy myopic policy) in a decoupled model, where the gain is zero, and a coupled investment model, where paying a worse first step to unlock the goal yields a positive gain. The right panel shows the cumulative expected free energy of every enumerated policy in the coupled model, with the greedy and optimal policies marked. The panel demonstrates that look-ahead value is a measured, model-dependent quantity, not a universal property of planning.

Continuous predictive coding as model-specific prediction-error descent

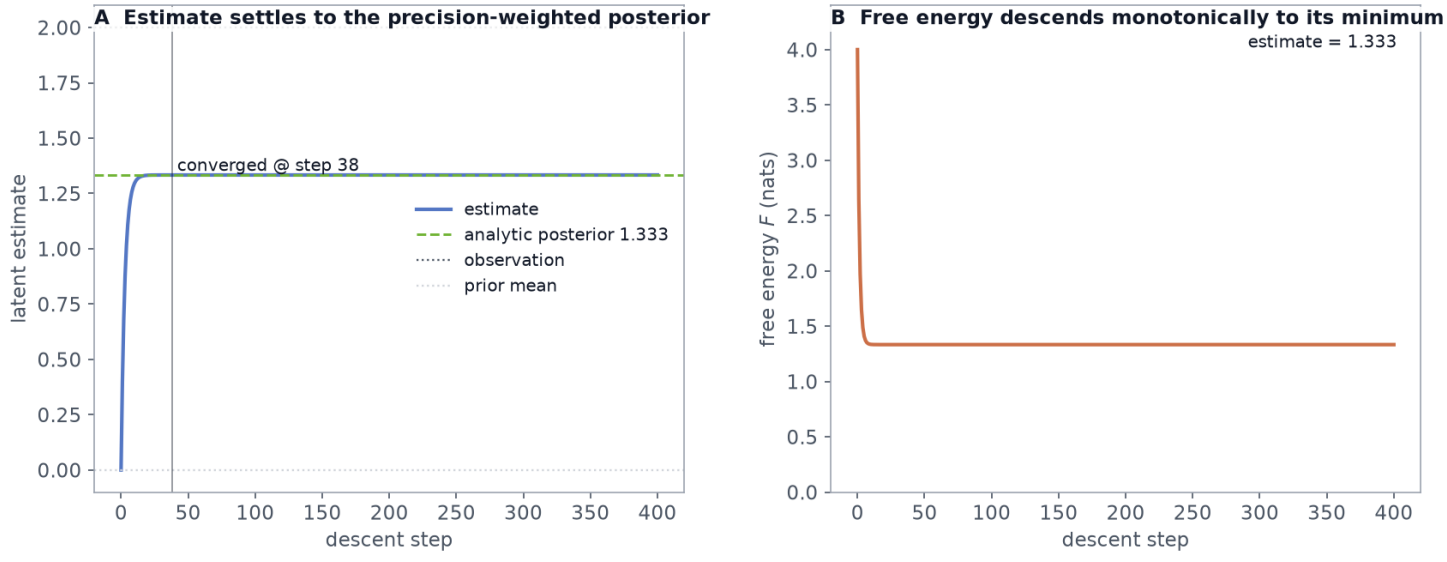
Continuous predictive coding as model-specific prediction-error descent I

The discrete examples above update beliefs in one shot; the continuous-state face of the Free Energy Principle instead *settles* a latent estimate by descending the free energy along precision-weighted prediction errors. On a single linear-Gaussian level – a vague unit-variance prior at zero meeting a more precise reading of 2 – the estimate relaxes from the prior mean to 1.33333, the exact analytic posterior mean, where the sensory prediction error (0.666667) and the prior prediction error (1.33333) balance once each is weighted by its precision.

fig. 6 shows the estimate converging to that fixed point while, under the stated objective and step size used in this simulation, the free energy decreases monotonically to its analytic minimum. The result is exact for the linear-Gaussian case the kernel tests pin against a closed form; it is a controlled teaching example, not a claim that prediction-error descent is exact for nonlinear or hierarchical models (Kalman 1960; Särkkä 2013; Buckley et al. 2017; Parr et al. 2022; Kataoka and Doya 2026).

Continuous predictive coding as model-specific prediction-error descent I

Predictive-coding parameter	Paper value
Prior mean and variance	$\mu_0 = 0,\ \sigma_0^2 = 1$
Observation and variance	$y = 2,\ \sigma_y^2 = 0.5$
Descent rate and steps	rate 0.1, maximum 400 steps
Analytic target	Gaussian posterior mean from the linear filtering update



Deterministic linear-Gaussian predictive-coding descent; the fixed point is the exact analytic posterior mean for this one model.

Continuous predictive coding as model-specific prediction-error descent II

Predictive-coding parameter	Paper value
Prior mean and variance	$\mu_0 = 0$, $\sigma_0^2 = 1$
Observation and variance	$y = 2$, $\sigma_y^2 = 0.5$
Descent rate and steps	rate 0.1, maximum 400 steps
Analytic target	Gaussian posterior mean from the linear filtering update

Figure 6: Continuous predictive-coding descent generated from the linear-Gaussian kernel. The left panel traces the latent estimate from the prior mean to the analytic precision-weighted posterior (dashed) as it balances the sensory and prior prediction errors; the right panel shows the free energy descending monotonically to its analytic minimum. The figure supports a bounded claim that prediction-error descent implements exact Gaussian inference for this one model, not a universal convergence guarantee for all generative models.